

Casa De Amparo - TRACE Parser vs. Field-Mapping

Field-by-field comparison of the energy-parser (src/engine/trace.ts + enrich.ts) against Casa_De_Amparo_Mapping.xlsx, with brief fix steps.

Generated 2026-06-15 • Source PDF: Casa De Amparo_v4 (TRACE 3D Plus 7.00, 725 pp).

0. How to read this report

The mapping spreadsheet is the intended derivation spec (which report / section / column each field comes from, and the formula). Its absolute numbers, however, come from a DIFFERENT model run than the uploaded PDF (mapping proposed total = 534,633 kBtu; this PDF = 742,871 kBtu). So the comparison below is about DERIVATION METHOD and SOURCE, not matching exact values.

Most fields already agree in method (see §3). Section 1 lists only the fields where our parser pulls from a different place or computes differently. Section 2 gives brief, low-risk steps to align them.

1. Field calculation differences

A. Envelope Thermal - NOT PARSED (largest gap)

Our parser reads wall AREAS and WWR from the "Lighting and Daylighting Summary" report, but never opens the "Envelope Summary" (Opaque Exterior Construction U-Factors, pp.408+) or the "Fenestration" table (glass U-Factor / SHGC / Visible Transmittance, pp.429+). All of the following therefore come out blank / 0:

- Wall U-Value** Mapping: Envelope Summary -> Opaque Exterior Construction, "U-Factor With Film". Ours: not parsed -> 0.
- Roof U-Value** Mapping: same section, Roof row (e.g. 0.026 in this PDF). Ours: not parsed -> 0.
- Exposed Floor U** Mapping: Envelope Summary floor construction. Ours: not parsed -> 0.
- Slab-on-Grade U** Mapping: Envelope Summary slab construction. Ours: not parsed -> 0.
- Glass U-Value** Mapping: Fenestration "U-Factor" (0.501). Ours: not parsed -> 0.
- Glass SHGC** Mapping: Fenestration "SHGC" (0.401). Ours: not parsed -> 0.
- Glass VLT** Mapping: Fenestration "Visible Transmittance" (0.442). Ours: not parsed -> 0.
- Glass LSG** Derived VLT/SHGC. Ours: 0 because VLT & SHGC are missing.
- Vert. Weighted U/R** Mapping: area-weighted wall+glazing (then $R=1/U$). Ours: 0 - vert_weighted_u never set for TRACE.
- Assembly U-Value** Mapping: area-weighted wall+roof. Ours: 0 - depends on wall/roof U which are missing.
- Infiltration Rate** Mapping: $ACH \times \text{volume} / 60$, summed, / facade area. Ours: hard-coded 0 in enrichRow.

B. Geometry ratios - wrong denominator

- Wall-to-Floor Ratio** Mapping: Gross Wall / CONDITIONED area ($11,729/13,652 = 0.859$). Ours: divides by gross/total area ($11,729/21,103 = 0.556$) - siteToRow uses gfa, not conditioned_floor_area.
- Roof-to-Floor Ratio** Mapping: Gross Roof / CONDITIONED area ($15,695/13,652 = 1.150$). Ours: / total area = 0.744. Same denominator bug.

C. Peak Loads - different physical basis

- Peak Heating (kBtuh)** Mapping: Table EAp2-5 "Heating Demand kW" x 3.412 (electric demand). Ours: Project Summary "Site Peak Heating Load (MBh)" = 410 (THERMAL load). Different quantities -> large gap (410 vs ~129 here).
- Peak Cooling (kBtuh)** Mapping: EAp2-5 "Cooling Demand kW" x 3.412. Ours: "Site Peak Cooling Load (tons)" x 12 = 300 (thermal). Different basis - decide which the workbook column intends.
- Peak Electrical (W)** Method matches (Peak Demand kW x 1000); ours reads Project Summary "Annual Peak Electrical Demand". Verify it is the same peak the mapping intends (LEED summary).

D. Economics - computed vs. read

- Total Energy Cost** Mapping: DIRECT read of Table EAp2-7 (Proposed 26,968 in this PDF). Ours: enrichRow COMPUTES cost = energy x virtual rate. It matches only when the model's virtual rate is used; with the finder/manual rate it diverges. We already parse the figure in parseEconomic() but never use it.

E. Carbon - factor value

Carbon Factor Mapping: 0.053 kg CO₂e/kBtu (~ 0.181 kg/kWh, CA grid). Ours: eGRID2022 CA = 0.21 kg/kWh -> 0.0615 kg/kBtu, ~16% higher. Config/reference difference, not a code bug.

F. System Efficiencies - not parsed for TRACE

Total Fan Power (kw) Mapping: sum of supply-fan kW from System Component Summary (proposed) / PRM fan power (baseline). Ours: total_fan_kw never set for TRACE -> 0.

DX / VRF / HP COP Mapping: manual ASHRAE 90.1 table references. Ours: 0 - expected to stay manual, but flag as a known gap.

G. Occupancy

Occupant Density Mapping: TOTAL building area / occupants (21,103/125 = 168.8 ft²/person). Ours: CONDITIONED area / occupants (13,652/125 = 109.2). Area-basis difference.

H. Per-direction WWR "Actual" (minor)

N/E/S/W WWR Actual Mapping: Window Opening Area / Gross Wall Area per direction. Ours: stores the "Gross Window-Wall Ratio" (nominal) value into the actual field. Values are ~identical here, but the formula differs. We already parse windowOpening per direction.

2. Steps to change (brief, low-risk)

All changes are in src/engine/trace.ts unless noted. Ordered by impact.

S1 - Parse the Envelope Summary + Fenestration (fixes all of A)

Add a page finder + two small parsers and set the row fields. The data layout (confirmed in this PDF):

Opaque Exterior Construction ... U-Factor With Film (BTU/h-ft²-°F)
Roof <name> <refl> <area> 0.026 ... -> row.roof_u_value
Wall <name> <refl> <area> 0.046 ... -> row.wall_u_value (area-wtd if many)
Fenestration ... U-Factor SHGC Visible Transmittance
90.1-13 Window ... 0.501 0.401 0.442 -> glass_u_value / glass_shgc / glass_vlt
- In parseTrace(): const ep = pages.find(p => p.text.includes('Opaque Exterior Construction')); const fe = pages.find(p => p.text.includes('Fenestration') && /U-Factor/.test(p.text));
- Regex per construction row: /b(Roof|Wall|Floor|Slab)\b[^\n]*?s([0-9.]+)s+[0-9.]+s+d+s*%/ - take the WITH-FILM column; average by gross area when several rows share a face.
- Fenestration: collect all /([0-9.]+)s+([0-9.]+)s+([0-9.]+)s+(?:No|Yes)/ triples after the header -> average -> glass_u_value, glass_shgc, glass_vlt.
- enrichRow already derives glass_lsg from VLT/SHGC. To fill vertical-weighted & assembly U, also set row.vert_weighted_u = (wallArea*wallU + winArea*glassU)/grossWallArea and compute assembly U = (wallArea*wallU + roofArea*roofU)/(wallArea+roofArea).

S2 - Fix wall/roof-to-floor denominator (fixes B)

In siteToRow(), the ratios are computed in the lighting block using gfa (gross). Move them after conditioned area is known, or just divide by conditioned_floor_area:

```
const denom = row.conditioned_floor_area || gfa;  
row.wall_to_floor_ratio = light.wallArea.total / denom;  
row.roof_to_floor_ratio = light.grossRoofArea / denom;  
row.envelope_to_floor_ratio = row.wall_to_floor_ratio + row.roof_to_floor_ratio;
```

S3 - Use the model's reported Total Energy Cost (fixes D)

We already parse Table EAp2-7 / Economic Comparison in parseEconomic() (utilityCost). Carry it onto the row and prefer it:

- In siteToRow(): match the alternative's economic row -> row.model_total_cost = econ.utilityCost.
- In enrichRow(): if data.model_total_cost > 0, use it for total_cost instead of energyxrate (keeps cost_intensity = model_total_cost / area).

S4 - Parse fan power (fixes F1)

Find the System Component Summary page; sum the supply-fan kW column -> row.supply_fan_kw / row.total_fan_kw (enrichRow already surfaces total_fan_kw / total_pump_kw).

S5 - Decide peak-load basis (C)

If the workbook columns mean ELECTRIC demand (mapping), parse Table EAp2-5 "Demand kW" per end use and use Heating/Cooling demand x 3.412. If they mean THERMAL load, keep the current Project Summary tons/MBh. Recommend confirming with the template owner before changing - cooling currently happens to match the mapping number.

S6 - Small one-liners (E, G, H)

- Carbon (E): set the project's grid factor to the intended value (~ 0.181 kg/kWh for CA) in the rate config, or switch carbon_method to Cambium; no code change needed.
- Occupant density (G): pick the area basis to match the template - change area in siteToRow's occ_density calc from conditioned to total_floor_area (or vice-versa).
- WWR actual (H): set row.<dir>_wwr_actual = light.windowOpening.<dir> / light.wallArea.<dir> x 100 (windowOpening is already parsed).

3. Fields that already match (no change needed)

Total energy & EUI (Site Consumption Grand Total = Table EAp2-6); electricity / gas; end-use breakdown (heating, cooling, fans, interior lighting & equipment); conditioned & total floor area; gross/N/E/S/W wall areas; gross roof & skylight; building & nominal WWR; ventilation / heating / cooling airflow (Project Summary = mapping values 1,550 / 4,082 / 8,767 cfm); LPD & EPD (0.45 / 0.69); electricity virtual rate; unmet heating/cooling hours (Project Summary agrees with Table EAp2-2: 105 / 547); peak electrical method; climate zone & weather file.